

A Low-Cost Visual Motion Data Glove as an Input Device to Interpret Human Hand Gestures

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Abstract — *Motion data gloves are frequently used input devices that interpret human hand gestures for applications such as virtual reality and human-computer interaction. However, commercial motion data gloves are too expensive for consumer use, and this has limited their popularity. This paper presents an inexpensive motion data glove to overcome this obstacle. To lower costs, we designed our glove to use single-channel video instead of expensive motion-sensing fibers or multi-channel video. Our visual motion data glove is composed of an inexpensive consumer glove with attached thin-bar-type optical indicators and a closed-form reconstruction algorithm that can overcome the common disadvantages of single-channel video approaches, i.e., occlusion and the need for inconvenient iterative reconstruction algorithms. Our low-cost visual motion data gloves are used to interpret human hand gestures, and the resulting performance is evaluated.*¹

Index Terms — *motion data glove, augmented reality, interpretation of human hand gestures, single-channel video.*

INTRODUCTION

The motion data glove is one of the most commonly used technologies for gesture-based human-computer interfaces. The motion data glove helps users to interface with the virtual world [1], and serves as an input device in non-verbal human-computer interaction. Commercial motion data gloves usually use expensive motion-sensing fibers and motion analyzers, and are consequently too costly for the consumer market [2]. Consequently, our research aim is to lower the cost of such equipment. A more cost-effective approach uses vision-based hardware. However, existing vision-based approaches are sensitive to lighting conditions and have a relatively low update rate [3]. To compensate for the disadvantages of both existing motion data gloves and other vision-based approaches, we propose a high-performance visual motion data glove. To lower the sensitivity to lighting conditions, we use appropriately-designed optical indicators attached to the motion data glove. Our visual motion data glove is manufactured by attaching optical indicators onto an inexpensive consumer glove and using an optical motion

capturing system for motion analysis. To improve update rates, we developed a motion analysis algorithm with lower computational cost.

Many optical motion capture systems have been commercialized, but they are usually based on multi-channel video. The multi-channel video approach needs many cameras and a large volume in which those cameras can be installed. These factors increase the cost of deployment. To lower the cost, we propose an optical motion capture system that uses only one camera, i.e., single-channel video. However, single-channel video approaches usually have several disadvantages, such as a requirement for complicated image analysis algorithms and a higher likelihood of occlusion. Commercial approaches to analyzing images from single-channel video aim to directly invert a complicated function that relates the parameters of the model to the observed image measurements [4], [5]. The difficulty inherent to this task is that the function in question is highly non-linear, which means that sophisticated iterative techniques are often required to identify plausible solutions. Moreover, it can be difficult to precisely characterize the set of possible solutions in terms of the set of allowable parameters [6].

To avoid highly complicated computation and obtain closed-form solutions, Taylor proposed a simple and effective algorithm for recovering the entire set of closed-form solutions by considering the foreshortening of the segments of the model in the image [6]. In this approach, the depth of each joint was recovered based on the prior knowledge of consecutive link lengths. Not surprisingly, this simple idea results in computational cost savings at the expense of some evaluation accuracy. Parallel to this approach, our work has aimed to develop an optical motion capture system with an image analysis algorithm that recovers the entire closed-form set of solutions. However, to achieve enhanced accuracy, we use a more refined model of each subject that is based on a series of revolving joint motions.

Usually single-channel video-based approaches suffer from more severe optical occlusion problems than multi-channel video. Accordingly, in a single-channel video approach, it is very important to design the optical indicators so as to minimize optical occlusions. Because human hands have multiple fingers, the occlusion problem is very severe. As an approach to avoid complicated optical occlusion due to finger movements, we propose an antenna-shaped optical indicator. Finally we apply our visual motion data glove to interpreting human hand gestures, and we validate its performance.

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II. Proposed Visual Motion Data Glove System

Fig. 1 shows a commercial motion data glove that incorporates motion-sensing fibers. Because motion-sensing fibers are very expensive and their analyzers are somewhat cumbersome, commercial motion data glove systems are too expensive for consumer usage.



Fig. 1. Commercial motion data glove that incorporates motion-sensing fibers [1].

To avoid expensive motion-sensing fibers we propose a visual motion data glove described in Fig. 2. In this figure, the joint position Y_{ij} and the optical indicator position Y'_{ij} are indicated by the circle and the black square, respectively. Because the relative positioning between the joint location and the optical indicator is known, the joint location can be computed easily given the coordinates of the optical indicator.

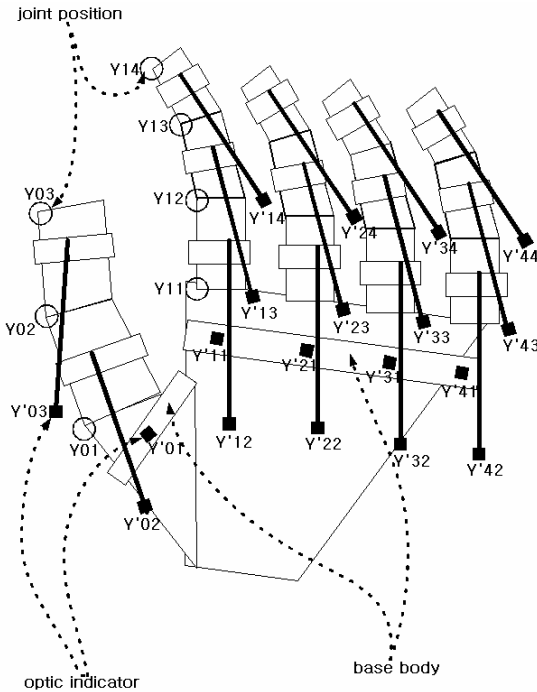


Fig. 2. Proposed visual motion data glove system.

As shown in Fig. 2, our visual motion data glove system is composed of a motion data glove and a corresponding visual motion analyzer. In our approach, the visual motion data glove is manufactured by attaching appropriately-designed optical indicators to an inexpensive consumer glove, and its visual

analyzer is composed of cameras and image analysis algorithms.

To lower the cost of the visual analyzer, we used single-channel video rather than multi-channel video. Compared with multi-channel video approaches, single-channel video has disadvantages such as a need for inconvenient iterative image analysis algorithms, and it suffers from occlusion problems.

Because human hands have multiple fingers, the occlusion problem is especially severe. To overcome the severely coupled occlusion problem, we propose long antenna-shaped optical indicators rather than the more normal point indicators, so that at least some parts of the indicators will be visible even when the fingers are completely hidden.

Iterative image analysis algorithms are somewhat inconvenient for real-time processing. Accordingly, we have attempted to design a simple closed-form algorithm. The proposed image analysis algorithm is described in detail in the following sections.

PROPOSED IMAGE ANALYSIS ALGORITHM OF A VISUAL ANALYZER FOR ROTATING JOINTS

We wished to design a visual analyzer algorithm that recovers a 3D pose, i.e., the joint angles of a human's hand, from image data generated by the optical indicators described in Fig. 2.

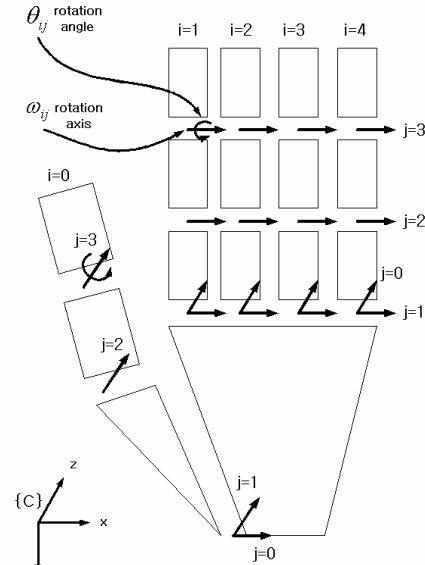


Fig. 3. Rotational joint model of a human hand.

In the design of the visual motion data glove described in Fig. 2, we use a model of hand motion as described in Fig. 3. In Fig. 3, a hand is modeled as a series of rotating joints. In this figure, $\{C\}$ denotes the camera frame, and i and j respectively denote a finger number and a joint level number. w_{ij} is the rotation axis vector of a rotating joint, and θ_{ij} is the rotation angle about w_{ij} . As shown in Fig. 2 the joint position Y_{ij} is easily computed from the measured point Y'_{ij} , i.e., the positions of the optical indicators attached to the visual motion data glove.

The joint point Y_{ij} is easily computed from the measured point Y'_{ij} . Because simultaneous analysis of the net total movement of a hand is very complicated, the computing time is high and it is difficult to find closed-form solutions. For real-time applications and convenience, we adopt the idea of dividing the overall movements of a hand into a series of simple sub-problems, each of which is related to a single joint. We first find the joint angle of one joint and then identify the joint angle of the next. In this way we compute the joint angles consecutively. We now design an algorithm for finding the joint angle of each joint as described in Fig. 3.

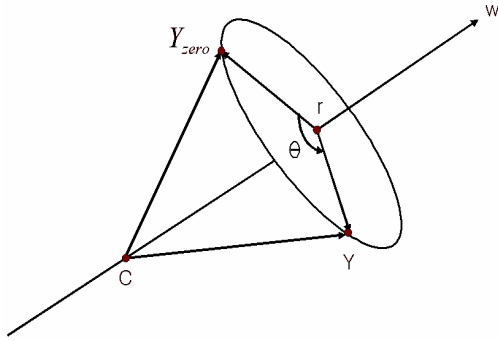


Fig. 4. Motion of a rotating joint. The point $Y_{zero} \in R^3$ moves to $Y \in R^3$ due to rotation about the axis $w \in R^3$ by angle $\theta \in R$. Here $r \in R^3$ is the center of rotation and $C \in R^3$ is a point on the joint axis w .

Fig. 4 shows a possible movement of a rotating joint. Suppose that a point $Y_{zero} \in R^3$ moves to $Y = zq \in R^3$ due to rotation about the axis $w \in R^3$ by angle $\theta \in R$ where $q \in R^2$ and $z \in R$ are respectively the image coordinate and the depth of the point Y . Then the following relation holds:

$$Y = \exp([\hat{w}]\theta)Y_{zero} \quad (1)$$

Here $\hat{w}$ is the unit vector in the direction of w . Let C be a point on the axis of rotation. Then the relation above can also be described as follows:

$$(Y - C) = \exp([\hat{w}]\theta)(Y_{zero} - C) \quad (2)$$

Using the Rodrigues formula [7]:

$$\exp([\hat{w}]\theta) = I + \sin(\theta)[\hat{w}] + (1 - \cos(\theta))[\hat{w}]^2$$

we can rewrite equation (2) as:

$$(Y - C) = k_1 + \cos(\theta)k_2 + \sin(\theta)k_3 \quad (3)$$

Here $k_1 = (I + [\hat{w}]^2)(Y_{zero} - C) \in R^3$,

$k_2 = -[\hat{w}]^2(Y_{zero} - C) \in R^3$, $k_3 = [\hat{w}](Y_{zero} - C) \in R^3$.

Substituting the relations:

$$\cos(\theta) = \frac{1-t^2}{1+t^2}$$

and

$$\sin(\theta) = \frac{2t}{1+t^2}$$

with $t = \tan(\theta/2)$ into equation (3) we obtain

$$Y = (C + k_1) + \left(\frac{1-t^2}{1+t^2}\right)k_2 + \left(\frac{2t}{1+t^2}\right)k_3 \quad (4)$$

Using a camera projection model, we obtain the corresponding image of the 3D point given in equation (4). Denote the projection mapping from Y to (q, z) as $\pi: R^3 \rightarrow R^2 \times R$, and its inverse projection mapping as $\pi^{-1}: R^2 \times R \rightarrow R^3$. Then the corresponding image point is as follows:

$$(q, z) = \pi(Y)$$

$$Y = \pi^{-1}(q, z) \quad (5)$$

Here Y is defined as in equation (4), and q is its corresponding image vector, of which the first and the second components are image coordinates, and the third component is set to 1. z is the depth corresponding to Y .

By combining equations (1) and (5), we obtain the following equation:

$$\pi^{-1}(q, z) = \exp([\hat{w}]\theta)Y_{zero} \quad (6)$$

Then, our estimation problem can be defined as the following optimization problem: Find $t \in R$ minimizing

$$J(t) = \|\tilde{q} - q(t)\|^2 \quad (7)$$

Here $q(t)$ is given in equations (4) and (5), and $\tilde{q}$ is a measurement value for q .

We obtain the solutions by solving the first order necessary conditions for optimality of equation (7) as follows:

$$\frac{\partial J}{\partial t} = 0 \quad (8)$$

Here J is defined as in equation (7).

Because analytical solutions of linear equations are well understood in the mathematical literature, we can obtain analytical solutions for equation (7) if equation (8) is described in terms of linear equations. Once $t \in R$ is found, we can compute the corresponding depth $z \in R$ using equation (6).

The results above can be summarized in the form of the following proposition.

Proposition 1: motion estimation for rotating joint

Suppose that a point Y_{zero} moves to $Y = \pi^{-1}(q, z)$ due to rotation about the axis w by angle θ where q and z are respectively the image coordinate and the depth of the point Y . For given Y_{zero} , q and w the optimal estimation of q and θ can be found by minimizing the objective function given in equation (7). If the first-order necessary condition in equation (8) is described in the form of linear equations, we can obtain analytical solutions using the following procedures:

Step 1: Find closed-form solutions satisfying equation (8).

Step 2: Compute θ and q by using the relation $t = \tan(\theta/2)$ and equation (5), respectively.

Step 3: Compute candidate values of z by using equation (6). By using the condition of $z > 0$ and applying limits to the joint angle we can choose a suitable result.

Many practical camera projection models satisfy the condition that the first-order expressions in equation (8) be described as linear equations. We now consider several examples of a representative camera projection model.

Case Study - Perspective Projection Model

As expressed in [8], the perspective projection model is one of the most popularly used camera models.

In the perspective projection model, equation (5) is described as follows:

$$q = \frac{f(Y - Y_t)}{e_3^T(Y - Y_t)}, \quad z = e_3^T Y \quad (9)$$

Here e_3 is the unit vector in the direction of the camera axis (for convenience, we take the Z-axis to be the camera axis in this paper), and Y_t is the optical center of the camera.

Substituting equation (4) into equation (9), we can describe the image coordinate of Y as the form:

$$q = \frac{m_1 t^2 + m_2 t + m_3}{n_1 t^2 + n_2 t + n_3} \quad (10)$$

Here $m_1 = f((C - Y_t) + k_1 - k_2) \in R^3$, $m_2 = f(2k_3) \in R^3$, $m_3 = f((C - Y_t) + k_1 + k_2) \in R^3$, $n_1 = e_3^T((C - Y_t) + k_1 - k_2) \in R$, $n_2 = e_3^T(2k_3) \in R$ and $n_3 = e_3^T((C - Y_t) + k_1 + k_2) \in R$.

Then the first-order necessary condition in equation (8) is described as the following equation:

$$\begin{aligned} \frac{\partial J}{\partial t} = & \left\{ \left(\frac{m_1 t^2 + m_2 t + m_3}{n_1 t^2 + n_2 t + n_3} \right) - \tilde{q} \right\}^T \times \\ & \left\{ \frac{(2m_1 t + m_2)(n_1 t^2 + n_2 t + n_3)}{(n_1 t^2 + n_2 t + n_3)} \right. \\ & \left. - \frac{(m_1 t^2 + m_2 t + m_3)(2n_1 t + n_2)}{(n_1 t^2 + n_2 t + n_3)^2} \right\} \\ = & 0 \end{aligned} \quad (11)$$

Rearranging this equation we obtain

$$(a_1 t^2 + a_2 t + a_3)^T (b_1 t^2 + b_2 t + b_3) = 0 \quad (12)$$

Here

$$\begin{aligned} a_1 &= (m_1 - n_1 \tilde{q}) \\ a_2 &= (m_2 - n_2 \tilde{q}) \\ a_3 &= (m_3 - n_3 \tilde{q}) \\ b_1 &= (m_1 n_2 - m_2 n_1) \\ b_2 &= 2(m_1 n_3 - m_3 n_1) \\ b_3 &= (m_2 n_3 - m_3 n_2) \end{aligned}$$

Expanding this equation leads to the following 4th order polynomial:

$$t^4 + c_1 t^3 + c_2 t^2 + c_3 t + c_4 = 0 \quad (13)$$

Here

$$\begin{aligned} c_1 &= (a_1^T b_2 + a_2^T b_1) / (a_1^T b_1) \\ c_2 &= (a_1^T b_3 + a_2^T b_2 + a_3^T b_1) / (a_1^T b_1) \\ c_3 &= (a_2^T b_3 + a_3^T b_2) / (a_1^T b_1) \\ c_4 &= (a_3^T b_3) / (a_1^T b_1) \end{aligned}$$

The polynomial above gives four closed-form solutions (refer to [9] for its solution).

Case Study - Affine Projection Model

One of the best-known camera models is the affine projection model [10]. In the affine projection model, equation (5) is described by the following equation:

$$q = AY + B, \quad z = e_3^T Y \quad (14)$$

Here

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 0 \end{bmatrix} \in R^{3 \times 3}, \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ 1 \end{bmatrix} \in R^3.$$

Substituting equation (4) into equation (14), we can describe the image coordinate of Y in the form:

$$q = \frac{m_1 t^2 + m_2 t + m_3}{n_1 t^2 + n_2 t + n_3} \quad (15)$$

Here $m_1 = A(C + k_1 - k_2) + B \in R^3$, $m_2 = 2Ak_3 \in R^3$, $m_3 = A(C + k_1 + k_2) + B \in R^3$, $n_1 = 1$, $n_2 = 0$, and $n_3 = 1$.

Because equation (15) has the same form as equation (10), solution procedures for this affine projection model are identical to those for the perspective projection model described in the previous section.

Performance Enhancement Possibilities

Movement of a joint is clearer in the video image when its motion is parallel to the image plane, i.e., the joint axis is parallel to the optical axis of the camera. (For convenience, we take the Z-axis to be the optical axis.) Clearer imagery leads to the better estimation of the 3D joint motion from the image data. Accordingly, we attempt to enhance performance using coordinate transforms that force the joint axis to become parallel to the Z-axis.

Suppose w is a joint axis that may not be parallel to Z-axis. The axis and angles of rotation from joint axis to Z-axis are respectively given by

$$\begin{aligned} w_r &= (a \times b) / \|a \times b\| \\ \theta_r &= \cos^{-1}((a^T b) / (\|a\| \|b\|)) \end{aligned} \quad (16)$$

Here $a = w$ and $b = [0, 0, 1]^T$.

Then the rotation matrix that transforms w to make it parallel to the Z-axis is

$$R_r = I + \sin(\theta_r)[w_r] + (1 - \cos(\theta_r))[w_r]^2 \quad (17)$$

We transform the coordinates of the relevant parameters using R_r .

$$w^r = R_r w$$

$$\theta^r = \theta$$

$$Y^r = R_r Y + P_r$$

$$\tilde{q}^r = (R_r \tilde{q}) / \{e_3^T (R_r \tilde{q})\} \quad (18)$$

Here P^r is appropriately chosen to prevent the Z-component of Y^r from becoming 0. If $Y \neq 0$ then $P_r = [0, 0, 0]^T$.

Using equation (13) and the relation $t^r = \tan(\theta^r / 2)$, we compute θ^r . The joint angle for w , i.e., θ is computed from θ^r for w^r using the following relation:

$$\theta = \theta^r \quad (19)$$

IX. PROPOSED ALGORITHM FOR VISUAL ANALYSIS OF HAND MOVEMENTS

In this section we describe how to apply the algorithm given in the previous section to interpret human body movements. The key idea is that we apply the algorithm given in the previous section to each joint of the human body in a round-robin fashion.

To demonstrate our algorithm we consider an example of a finger as described in Fig. 5. $Y_{1zero}, Y_{2zero}, Y_{3zero}, Y_{4zero} \in R^3$ are the coordinates of the joints in a defined zero configuration, which means the configuration in which all joint angles are zero. These points respectively move to $Y_1, Y_2, Y_3, Y_4 \in R^3$ due to rotation of the joints by $\theta_1, \theta_2, \theta_3$. Let the axes of the joints at the reference configuration be $w_{1zero}, w_{2zero}, w_{3zero} \in R^3$. Then Y_1, Y_2, Y_3 and Y_4 can be described by the product-of-exponentials formula given below.

$$\begin{aligned} Y_1 &= Y_{1zero} \\ Y_2 &= \exp([\hat{w}_{1zero}] \theta_1) Y_{2zero} \\ Y_3 &= \exp([\hat{w}_{2zero}] \theta_2) \exp([\hat{w}_{1zero}] \theta_1) Y_{3zero} \\ Y_4 &= \exp([\hat{w}_{3zero}] \theta_3) \exp([\hat{w}_{2zero}] \theta_2) \exp([\hat{w}_{1zero}] \theta_1) Y_{4zero} \end{aligned} \quad (20)$$

Here $w_{1zero}, w_{2zero}, w_{3zero}, Y_{1zero}, Y_{2zero}, Y_{3zero}, Y_{4zero} \in R^3$ can be easily obtained by considering the configuration in Fig. 3 and by using prior knowledge of the distances between joints.

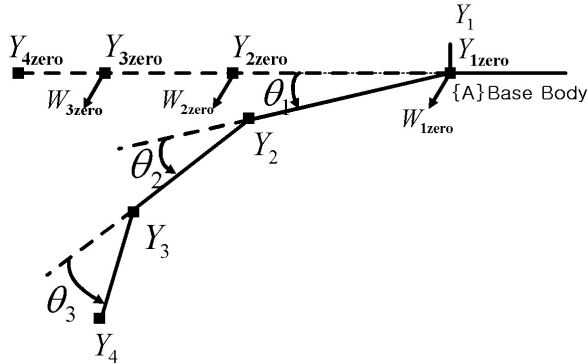


Fig. 5. Motion of a finger with respect to a defined zero configuration.

The posture of the human body can be estimated as follows.

Proposition 2: Closed-form solution for estimating a human hand's posture described in Fig. 5:

Step 1: Choose z_1 , and compute $Y_1 = \pi^{-1}(q_1, z_1)$.

Step 2: Consistent with Fig. 3, we choose $w_{1zero}, w_{2zero}, w_{3zero}$

as unit vectors in the direction of X-axis. Considering the configuration in Fig. 3 and prior knowledge of the distances between joints, we establish $Y_{1zero}, Y_{2zero}, Y_{3zero}, Y_{4zero} \in R^3$ with $Y_{1zero} = Y_1$.

Step 3: For $i=1$ to $i=3$, iterate repeatedly through Step 4 and Step 5.

Step 4: Let $C_i = Y_i$. Using the algorithm presented in Proposition 1, find θ_i, z_{i+1} and q_{i+1} that minimize

$$J_i(t) = \|\tilde{q}_{i+1} - q_{i+1}(t)\|^2.$$

And compute z_{i+1} using the equation:

$$\pi^{-1}(q_{i+1}, z_{i+1}) = \exp([\hat{w}_i] \theta_i) Y_{izero}$$

Step 5: Compute Y_{i+1} using equation (20).

Step 6: Repeat Step 1 to Step 5 for each finger of the hand.

Using the procedures above we obtain a mathematical model of hand posture. In Step 1, if we know the exact value of z_1 , we can obtain the absolute position of the hand. One simple way to do this is to choose z_1 by comparing the projected image size of the corresponding marker with its real size. A more accurate approach is to apply Han's algorithms presented in [11], [12] to joint points on the base body and use prior knowledge about the distances between each of them.

X. RESULTS

This section first shows an example implementation of the proposed visual motion data glove, and then verifies the performance of the proposed visual analyzer algorithm.

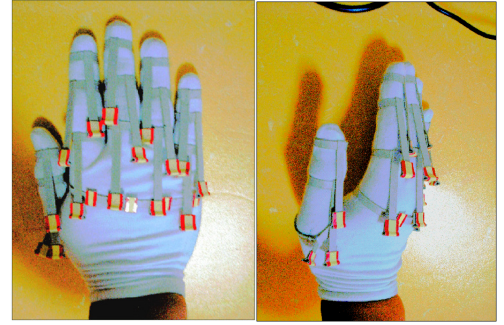


Fig. 6. An example implementation of our proposed visual motion data glove.

Fig. 6 shows an example implementation of the proposed visual motion data glove. We first capture a hand's image, and then we detect the optical indicators attached to the hand by color-matching. By applying the proposed visual analyzer algorithm to the images of those detected indicators, we recover its 3D positions and thus joint angles. To capture the hand's motion we use a monocular camera whose intrinsic parameters are known and assumed to be unchanged throughout the sequence. The camera is installed on a stationary platform. The distance moved is about 0.34 meters, and the image size is 512 x 384 pixels. Our experimental scenarios are composed of 3 classes, namely, button push, number and OK sign. The scenarios are computed using MATLAB on a Pentium PC with a clock speed of 1.20 GHz and 540 MB RAM.

Button-Push

As a first scenario we consider the button-push problem described in Fig. 7, Fig. 8 and Fig. 9.

In Fig. 7, Fig. 8 and Fig. 9, the left and the right figures show the input images of the ready state, the left-button-pushed state and the right-button-pushed state for the button-push scenario and their reconstructed 3D models. The solid

and dotted lines denote the reconstructed 3D models obtained by the proposed approach and by manual joint angle measurement, respectively.

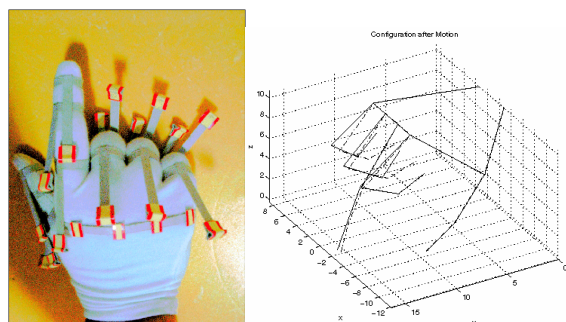


Fig. 7. Ready state of button-push scenario and its reconstructed image. The left figure and the right figure show the ready state of the button-push scenario and its reconstructed image using the proposed approach, respectively.

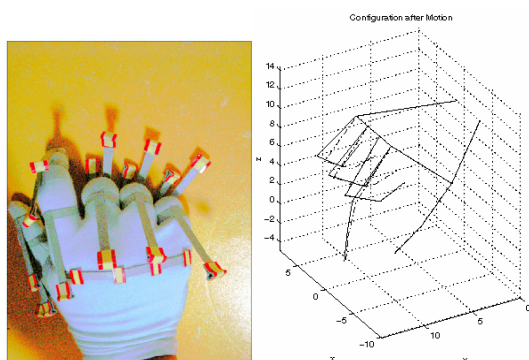


Fig. 8. Left-button-pushed state of button-push scenario and its reconstructed image. The left figure and the right figure show the left-button-pushed state of the button-push scenario and its reconstructed image using the proposed approach, respectively.

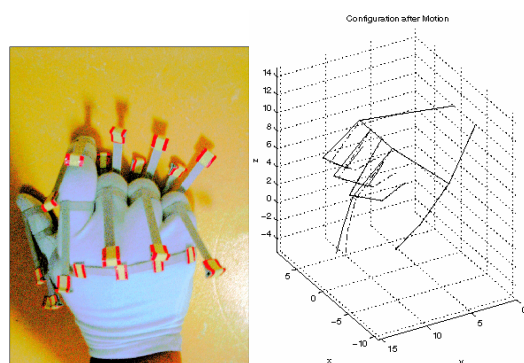


Fig. 9. Right-button-pushed state of button-push scenario and its reconstructed image. The left figure and the right figure show the right-button-pushed state of the button-push scenario and its reconstructed image using the proposed approach, respectively.

Table I shows average error ratios of the reconstructed joint angles to the measured joint angles in Fig. 7, Fig. 8 and Fig. 9. In this figure we compare the results of the proposed algorithm with those of two contemporary estimation algorithms for 3D finger motion proposed by Diffranco [13] and Lee [14]. The algorithms are a nonlinear iterative type of

process, and an analytical solution type algorithm, respectively. Table I confirms that the proposed approach shows much more accurate reconstruction results than the existing analytical approach. The reconstruction accuracy of the proposed approach is nearly the same as the existing nonlinear iterative approach.

TABLE I
AVERAGE ERROR RATIO OF RECONSTRUCTED JOINT ANGLES IN BUTTON-PUSH SCENARIO

	Our	Lee	Diffranco
Ready	0.1509	0.3325	0.1253
Left-push	0.1791	0.4137	0.1419
Right-push	0.1510	0.4144	0.1120
Average	0.1603	0.3869	0.1264

Table II shows compute time (in seconds) required to calculate the reconstructed joint angles in Fig. 7, Fig. 8 and Fig. 9.

From this table, it is clear that the proposed approach requires much less compute time than the existing nonlinear iterative approach. The computing time of the proposed approach is nearly the same as the existing analytical approach. The existing nonlinear iterative approach can process only two images per second, which is too slow for true real-time processing. In contrast, the proposed and the analytical approaches can process about 4 times faster than the existing nonlinear iterative approach.

TABLE II
COMPUTING TIME TO CALCULATE RECONSTRUCTED JOINT ANGLES IN BUTTON-PUSH SCENARIO

	Our	Lee	Diffranco
Ready	0.1226	0.1350	0.4917
Left-push	0.1263	0.1264	0.4832
Right-push	0.1214	0.1387	0.4794
Average	0.1234	0.1334	0.4848

In summary, Table I and Table II show two key facts. First, there is a trade-off problem between reconstruction accuracy and computing time. Second, the proposed approach has as high a reconstruction accuracy as the existing nonlinear iterative approach but a low computing time that is comparable to the existing analytical approach.

Numbers

As a second scenario, we considered numerical numbering as described in Fig. 10, Fig. 11 and Fig. 12.

In Fig. 10, Fig. 11 and Fig. 12, the left and the right figures show the input images of the number one, the number two and the number three states of the numerical numbering scenario and their reconstructed 3D models. The solid and dotted lines denote the reconstructed 3D models obtained by the proposed approach and by manual joint angle measurement, respectively.

Table III shows average error ratios of the reconstructed joint angles to the measured joint angles in Fig. 10, Fig. 11 and Fig. 12. In this table, we can see that the proposed approach yields much more accurate reconstruction results than the existing analytical approach. The reconstruction accuracy of the proposed approach is nearly the same as the existing nonlinear iterative approach.

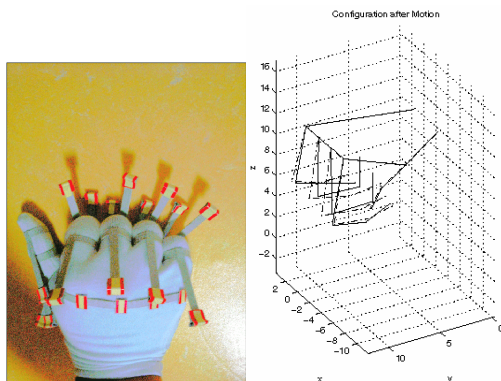


Fig. 10. Number one scenario and its reconstructed image. The left figure and the right figure show the captured image of the number one scenario and its reconstructed image using the proposed approach, respectively.

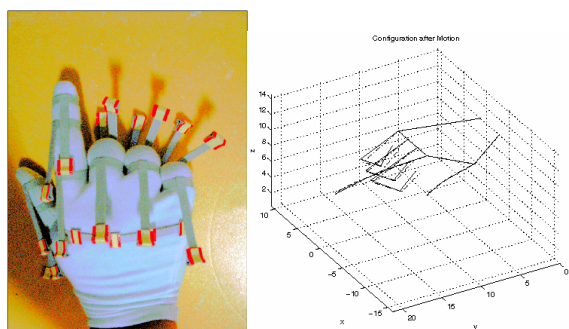


Fig. 11. Number two scenario and its reconstructed image. The left figure and the right figure show the captured image of the number two scenario and its reconstructed image using the proposed approach, respectively.

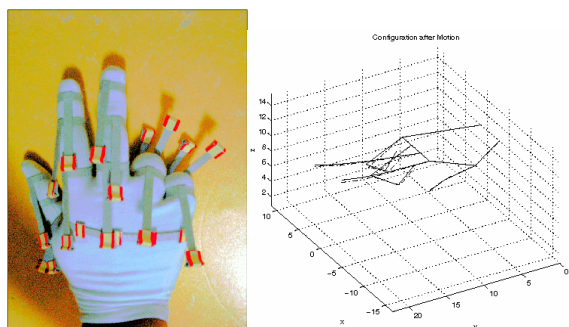


Fig. 12. Number three scenario and its reconstructed image. The left figure and the right figure show the captured image of the number three scenario and its reconstructed image using the proposed approach, respectively.

Table IV shows computing time (in seconds) required to calculate the reconstructed joint angles in Fig. 10, Fig. 11 and

Fig. 12. The proposed approach requires much less computing time than the nonlinear iterative approach. The computing time needed for the proposed approach is nearly the same as that required for the existing analytical approach.

In summary, Table III and Table IV show two key facts. First, there is a trade-off problem between reconstruction accuracy and computing time. Second, the proposed approach has as high reconstruction accuracy as the existing nonlinear iterative approach but as low computing time as the existing analytical approach.

TABLE III
AVERAGE ERROR RATIOS OF RECONSTRUCTED JOINT ANGLES FOR THE NUMBER SCENARIO

	Our	Lee	Difranco
One	0.6822	0.9491	0.5461
Two	0.2222	0.3390	0.2038
Three	0.2813	0.3821	0.2313
Average	0.3952	0.5567	0.3271

TABLE IV
COMPUTING TIME FOR CALCULATING RECONSTRUCTED JOINT ANGLES IN THE NUMBER SCENARIO

	Our	Lee	Difranco
One	0.1242	0.1366	0.5685
Two	0.1226	0.1378	0.4254
Three	0.1255	0.1406	0.4267
Average	0.1241	0.1383	0.4735

OK Sign

As a third scenario we considered the OK sign described in Fig. 13. In Fig. 13, the left and the right figures show the captured image of the OK sign scenario and its reconstructed 3D models as generated by manually measuring the joint angles.

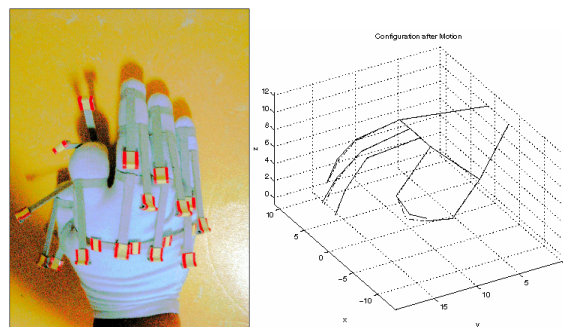


Fig. 13. OK sign scenario (left) and its 3D model (right).

To assess performance, we designed experiments in which we could produce the required input images by projecting the manually measured 3D models onto the image plane and adding measurement noise to the projected joint image positions. In addition we added measurement errors to the

initial joint depths. The measurement noise and errors were adjusted consistent with the experiment conditions that we wished to consider.

Table V, Table VI and Table VII give average error ratios (computing times shown in parentheses) of the reconstructed joint angles to the manually measured joint angles in the scissors scenario when the error ratio in the base joint was 0.05, 0.10, 0.15, respectively. The columns of each table indicate different image noise conditions. The units of image noise and computing time are pixels and seconds, respectively.

TABLE V
AVERAGE ERROR RATIO (COMPUTING TIME IN SECONDS) OF
RECONSTRUCTED JOINT ANGLES IN THE OK SIGN SCENARIO WHEN THE
ERROR RATIO IN THE BASE JOINT IS 0.05.

Image noise	Our	Lee	Diffranco
0.5	0.1352 (0.1266)	0.3491 (0.1325)	0.1010 (0.4606)
1.0	0.2039 (0.1250)	0.3934 (0.1329)	0.1705 (0.4869)
1.5	0.2851 (0.1264)	0.4409 (0.1287)	0.3647 (0.5472)
2.0	0.3439 (0.1302)	0.4920 (0.1339)	0.3446 (0.5647)
Average	0.2420(0.12 71)	0.4189(0.13 20)	0.2452(0.51 49)

TABLE VI
AVERAGE ERROR RATIO (COMPUTING TIME IN SECONDS) OF
RECONSTRUCTED JOINT ANGLES IN THE OK SIGN SCENARIO WHEN THE
ERROR RATIO IN THE BASE JOINT IS 0.10.

Image noise	Our	Lee	Diffranco
0.5	0.2142 (0.1265)	0.4989 (0.1325)	0.1996 (0.5810)
1.0	0.2543 (0.1279)	0.5339 (0.1287)	0.2275 (0.5924)
1.5	0.2771 (0.1188)	0.5745 (0.1325)	0.2558 (0.5846)
2.0	0.3424 (0.1262)	0.6180 (0.1288)	0.3860 (0.5969)
Average	0.2720(0.12 49)	0.5563(0.13 06)	0.2673(0.58 89)

TABLE VII
AVERAGE ERROR RATIO (COMPUTING TIME IN SECONDS) OF
RECONSTRUCTED JOINT ANGLES IN THE OK SIGN SCENARIO WHEN THE
ERROR RATIO IN THE BASE JOINT IS 0.15.

Image noise	Our	Lee	Diffranco
0.5	0.2923 (0.1227)	0.6326 (0.1266)	0.2833 (0.6948)
1.0	0.3132 (0.1302)	0.6600 (0.1300)	0.3101 (0.6708)
1.5	0.3529 (0.1340)	0.6967 (0.1313)	0.3361 (0.7098)
2.0	0.3754 (0.1238)	0.7334 (0.1288)	0.4152 (0.6721)
Average	0.3335(0.12 77)	0.6807(0.12 92)	0.3362(0.68 69)

From these tables, it is clear that the proposed approach shows much more accurate reconstruction results than the existing analytical approach. The reconstruction accuracy of the proposed approach is nearly the same as that of the existing nonlinear iterative approach. In addition, the reconstruction accuracy of the proposed approach is less

sensitive to the error ratio in the base joint than the existing analytical approach. This is to be expected because the existing analytical approach is fundamentally based on 3D joint position, so its sensitivity to 3D joint position is inevitable. In contrast, our proposed approach is based mainly on 3D joint angles, which are more closely correlated with relative joint positions than with absolute joint positions. For this reason, the proposed approach is less sensitive to absolute 3D joint positions. Accordingly, it is more robust to measurement errors in the initial joint depth. Considering the fact that the initial joint depth is difficult to measure accurately and joint angles are more important than absolute joint positions in estimating hand pose, this feature of the proposed approach is very favorable.

In terms of computing time, the three tables show that the proposed approach requires substantially less computing time than the existing nonlinear iterative approach. The computing time of the proposed approach is nearly the same as that of the existing analytical approach. The computing time of the existing nonlinear iterative approach has a tendency to increase as the image noise or error ratios in the base joints increase, and this makes it difficult to predefine the number of processed images per second. In contrast, the proposed approach has nearly constant computing times that are robust to image noise and error ratios. This is one of the main advantages of analytical approaches as compared with nonlinear iterative approaches.

X. CONCLUSION

We developed a visual motion data glove and showed results from gesture recognition using this glove. Commercial motion data gloves usually use high-cost motion-sensing fibers to acquire hand motion data, but in this paper we adopted a single-channel video system to reduce cost. Our visual motion data glove uses thin-bar-shaped optical indicators to mitigate occlusion-related problems. For real-time performance and convenience we designed the 3D hand motion estimation algorithm as a closed analytical form instead of using a nonlinear iterative approach. Performance evaluations verified that our algorithm has a lower and more consistent computing time relative to nonlinear iterative-type algorithms. The reduced computing time is useful for real-time applications, and the more consistent computing time makes it possible to predefine the number of processed image frames per second. In contrast with many other closed-form analytical algorithms, our chosen algorithm estimates hand motion via local optimization of joint angles instead of by direct derivation from 3D joint positions. Performance evaluations verified that our algorithms have lower estimation errors than those existing closed-form analytical approaches. Nonetheless, our algorithms maintain very nearly the same computing performance. Furthermore, we also verified that estimation errors associated with our algorithm are less sensitive to measurement errors in the initial joint depths. This feature is very desirable, considering the fact that it is very difficult to accurately measure the depth of the initial joint.

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BIOGRAPHY



Youngmo Han received a B.S. degree in physics education, B.E. degree in control and instrumentation, M.E. degree in electric engineering, and a Ph.D. degree in Mechanical and Aerospace Engineering, all from Seoul National University, Seoul, Korea, in 1992, 1995, 1998, and 2002, respectively. In his M.E., he studied signal processing and computer-aided VLSI design. In his Ph.D., he studied computer vision, 3D Pose recognition of human body images and robot systems. From 2002 to 2003, he was with the UAV group of Sejong-Rockheed Martin Research Center as a researcher and a research professor in Seoul, Korea. From 2004 to 2006 he was a full-time research lecturer and a research professor with the Department of Information Electronics Engineering, Ewha Womans University, Seoul, Korea. Since 2006, he has been a full-time lecturer and an assistant professor with the Department of Computer Engineering, Hanyang Cyber University, Seoul, Korea. Since 2007, he has been a Deputy Director General and an Honorary Director General with the International Biographical Centre, England. Since 2007, he has been a Vice-President of World Congress of Arts, Science and Communications, England. Since 2009, he has been a Deputy Governor with the American Biographical Institute Research Association, USA. Since 2009, he has been a Vice-Chancellor of the World Academy, USA. His research interests are in applications of computer vision to multimedia and biological image recognition, software and algorithm of mobile and robot embedded systems, and their VLSI and Embedded SOC implementations, human-computer visual interfaces, and fusion-type integrated approaches for information technology. He has written more than 10 journal papers at major international journals on these research themes, including 5 IEEE Transactions and Elsevier journals. It is of note that he has written full papers in a variety of research areas, introducing new fusion-type integrated approaches. Because of his active publications, he has attracted attention of many international associations. Many international journal associations have invited him to publish papers. His biography and research achievements were introduced about ten times by Marquis Who's Who, USA, and about 20 times by the International Biographical Centre, England and several times by the American Biographical Institute, USA. Especially, he was nominated for IBC Hall of Fame (2007), Top 100 engineers of the world (2008) and Top 100 scientists of the world (2009) by International Biographical Centre, and ABI Hall of Fame (2009) by American Biographical Institute. He was also awarded a 'Lifetime Achievement Award' by the World Congress of Arts, Science and Communications, England.